

MALLIKARJUN BHUSNOOR

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EDUCATION

University of Illinois at Urbana Champaign
MS in Information Management | GPA : 4.0

Champaign, USA
August 2024 – May 2026

KJ Somaiya College of Engineering, Mumbai University
Bachelor of Technology in Electronics and Telecommunication | GPA : 3.8

Mumbai, India
August 2019 – May 2023

PROFESSIONAL EXPERIENCE

Healthy Regions & Policies Lab (HeRoP), Champaign, USA
Data Scientist (Research Assistant)

September 2025 – Present

- Independently engineered **RAAM** and **2SFCA** spatial accessibility models (**PySAL**, **GeoPandas**) across 74K census tracts to quantify opioid treatment access for **13 JCOIN** institutions under NIH's \$150M+ HEAL Initiative.
- Implemented IDW (scipy) and Ordinary Kriging (pykrige, spherical variogram) interpolating PM2.5 from 280 sensors into weekly surfaces across 801 Chicago census tracts with holdout correlation validation.
- Developed **R spatial pipelines** (**sf**, **dplyr**, **tmap**) computing nearest-facility distances and 30-min drive-time access to 5 opioid treatment modalities across all 50 states as OEPS health equity.
- Architected a containerized **ETL** pipeline (**Docker**, **GitHub Actions**, **AWS S3**) ingesting PM2.5 and **AQI** from 280 Chicago sensors via Clarity **REST API** with R **QA/QC** delivering **Parquet** datasets to the ChiVes dashboard.

Rail Transportation and Engineering Center (RailTEC), Champaign, USA
Data Scientist (Research Assistant)

January 2025 – September 2025

- Automated a **Random Forest** derailment **classification** pipeline (**Airflow**, **Python**) on **2M+** records achieving **20% F1** improvement and **40%** runtime reduction through **feature selection** and **multicollinearity** removal.
- Quantified rail incident drivers across **400K+** records via **ANOVA**, **OLS**, and **correlation analysis**, isolating **12** significant track-condition, weather, and operational predictors for **Random Forest feature selection**.
- Engineered a **GEE-SRTM** pipeline computing signed grade and curvature from **30m** elevation samples across 100K+ track vertices, replacing manual **GIS processing** and reducing runtime from 1 week to **2** hours.

Emkay Global Financial Services Limited, Mumbai, India
Business Analyst

June 2023 - June 2024

- Refined **real-time trading ETL** logic across 20,000+ records and **78 bid/spread variables**, eliminating redundant transformations and restructuring validation checks to improve execution speed by **15%**.
- Created Tableau dashboards **tracking** algorithm performance across 2.16B trades, converting Sharpe ratio, volatility, and execution metrics into trading insights contributing to a **12%** profitability improvement.
- Streamlined recurring **portfolio reporting** by standardizing **data pulls**, **validation checks**, and **stakeholder-ready** outputs, reducing turnaround from 3 hours to minutes and cutting vendor-related downtime by **~20%**.

ACADEMIC PROJECTS

The Last Mile: Predicting Opioid Mortality in Pharmacy Deserts | Python, XGBoost, SHAP, MySQL, OSMnx, Tableau | [GitHub](#)

- Constructed a **spatial ML** pipeline predicting opioid mortality across 3,059 counties with 12 features: LISA cluster encodings, **ArcGIS Gi*** scores, and **OSMnx** road distances to 40,135 pharmacies via **SLURM** HPC.
- Trained **XGBoost** (**R² = 0.52**) with SHAP identifying spatial cluster type as dominant mortality predictor (**4.5 mean SHAP** units) and estimating **5.7 deaths/100K** reduction if pharmacy deserts were eliminated.

Grocery Sales Forecasting - Corporación Favorita | Python (XGBoost, CatBoost, scikit-learn), AWS SageMaker

- Evaluated **XGBoost**, LightGBM, CatBoost, and **Random Forest** in a 4-phase ablation on **41.5M records** via **AWS SageMaker**, engineering **32** leakage-guarded temporal features achieving **35% SMAPE** and **19% RMSE** improvement.
- Isolated **day_of_week** as the #1 demand driver across 218K store-item series and 33 product families, revealing weekly **behavioral** rhythm outperforms **9** engineered **lag** features consistently across all 4 models.

Spatial Modeling of Healthcare Access Disparities | Python (PySAL, GeoPandas, scikit-learn), Random Forest, OLS

- Identified **SAR** as the only model with positive **R² (0.279)** across **OLS**, **Ridge**, and **Random Forest** on county-level healthcare access, reducing residual spatial autocorrelation (**Moran's I**) from **0.51** to **-0.05**.
- Derived a Health Access Risk Index (HARI) combining SVI and access deficit scores across 3,100+ counties (PySAL, GeoPandas) to surface high-risk provider shortage clusters nationally.

PUBLICATIONS

Bhusnoor, M. S. et al. "Assessment of Performance and Emission Characteristics of Compression Ignition Engine Using Diesel and Biodiesel Blends." *Journal of Environmental Informatics Letters*, Vol. 10(1), 2023. (Scopus Indexed, First Author) - doi.org/10.3808/jeil.202300109

TECHNICAL SKILLS

- Languages / Libraries** : Python , R , Pandas, NumPy, Scikit-learn, XGBoost, CatBoost, LightGBM, TensorFlow, PySAL, GeoPandas
- ML / Modeling**: Random Forest, Gradient Boosting, Spatial Regression (SAR), Spatial Accessibility (RAAM, 2SFCA), SHAP, CNN
- Data Engineering & Cloud**: Docker, GitHub Actions, Airflow, AWS (S3, SageMaker), SLURM HPC, MySQL, SQL Server, Git/GitHub
- Spatial Analysis** : ArcGIS Pro, Google Earth Engine, GeoDa
- Visualization**: Tableau, Power BI (Power Query, DAX), Excel (Advanced)